

Owen Porteous

(615) 938-4166 | oporteous@vols.utk.edu | linkedin.com/in/OwenPorteous

PROFESSIONAL SUMMARY

Dual Aerospace Engineering and MBA graduate student at the University of Tennessee, Knoxville, with hands-on experience in launch vehicle design, propulsion, flight simulation, and guidance and navigation. Led a 13-member senior design team through the design, manufacturing, testing, and launch of a custom hybrid rocket engineered for 45,000+ feet, with additional experience developing 6-DOF simulation and Apollo-based flight software. Seeking a space industry internship to apply aerospace systems knowledge, technical problem-solving, and cross-functional leadership to launch vehicle, spacecraft, or mission development.

EDUCATION

University of Tennessee

Tickle College of Engineering - *Master of Science in Aerospace Engineering*

Haslam College of Business - *Master of Business Administration*

Tickle College of Engineering - *Bachelor of Science in Aerospace Engineering*

Knoxville, TN

May 2028

May 2027

May 2026

PROFESSIONAL EXPERIENCE

UT SENIOR DESIGN, *Technical Team Lead*, Knoxville, TN

August 2025 - June 2026

- Team Lead of 13 students across 5 sub-teams, planning weekly meetings, maintaining technical understanding across each sub-team, and making final design decisions
- Built and tested a custom high-powered hybrid rocket designed for 45,000+ ft altitude and competed at FAR-OUT in the Mojave Desert in the 20k'-50k' category
- Diagnosed and solved the previous team's failures by combining the post-combustion chamber and nozzle into one part
- Evaluated our own failed test fire, and within a one-week turnaround, conducted the first successful full-duration static fire for this rocket
- Oversaw design and machining of custom aluminum and graphite components, and assisted in the modeling, design, and 3D printing of the electronics housing
- Coordinated around a budget of \$6,000 and raised an additional \$4,000 through crowdfunding

MCLAURIN AEROSPACE, *Flight Dynamics Research Intern*, Knoxville, TN

June 2024 - August 2025

- Extended development of a MATLAB-based 6 DOF launch vehicle simulation, conducted extensive research into the Apollo Lunar Module descent stage, and spearheaded first real vehicle with testable flight data
- Implemented Apollo navigation and guidance as brand-new flight software by taking sensor data and converting it into acceleration commands for the engine through coordinate transformations, onboard trajectory analysis, and numerical integration
- Added key infrastructure such as throttleable engine capabilities using an exponential ramp model for throttle response lag, paving the way for control systems after guidance output

SKILLS

Software & Tools: MATLAB, Python, Excel, PowerPoint, Git/GitHub, Claude Code

Technical Skills: Flight software, Guidance & navigation, Coordinate frame transformations, 6 DOF simulation, Trajectory analysis, PID control, Hybrid propulsion, Static fire testing, Propulsion system assembly, Failure diagnosis, Problem solving, Physics, Math

Personal Skills: Creative solutions, Collaborative, Coaching, Leadership, Project Development

PROFESSIONAL ACCOMPLISHMENTS

University of Tennessee Dean's List

Peter Barile Sr. Design Competition Scholarship Award

August 2022 - May 2026

August 2023